

SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID – SIH26103

Problem Statement Title – Use case on web-based integrated project-monitoring platform

Theme – Smart Automation

PS Category – Software

Team ID –

Team Name – BUBBLESORT



PRAKALP-DRISHTI — calibrated early warning for PAIMANA

IDEA TITLE

PRAKALP-DRISHTI — tells IPMD which PAIMANA projects will slip, when, and how sure we are



Real-world issue

PAIMANA records a delay **after** it happens. **1,988 of 2,103** projects are still running — nobody can say which one slips next.



Why it matters

1,981 ongoing projects; cost already up from **₹37.13 to ₹42.78 lakh crore**. Every month of warning is time to act.



Solution

A monthly add-on to IPMD's report. For every ongoing project: a **likely finish window**, a **cost-overflow estimate**, and a **ranked list of what to look at first** — with the reason.



Prototype

Live ▶ prakalp-drishhti.vercel.app
Code ▶ github.com/amey-bauchkar/Prakalp-Drishti-

Proposed Solution (Describe your Idea/Solution/Prototype)

617185 · Kurnool-IV REZ transmission (POWERGRID) · ₹5,550 Cr
25% built · sanctioned Mar 2025 · **not yet late by PAIMANA's own dates**

PAIMANA's own dates (red):
original 23 Mar 2027 · revised 30 Sep 2027

22% chance it finishes by PAIMANA's Sep 2027 date



Read it like this: most likely finish Nov 2028; the band holds 10–95% of outcomes. It is wide because the project is 25% built in 15 months on a 24-month plan, and this sector has few finished projects to learn from.

Tested on 596 held-out projects: the official revised date fell inside the band 79% of the time (51% before calibration).

How it addresses the problem

- a** **Statistical and predictive models** → three built (survival model for time; OLS and boosting for cost), tested on projects newer than the training data
- b** **Does AI beat conventional statistics?** → boosting beats the sector average by 12% but OLS by only 2.6%; the real gain is calibration — window hit-rate 51% → 79%
- c** **Extra variables beyond CUF fields?** → rain, prices, elections and seven more did not help; land, clearance and milestone fields would — our ask to MoSPI

Outcomes covered: a b c d e g h i · f partly

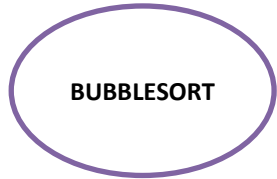
Detailed explanation of the proposed solution

- Finish window** — a likely finish date with a best-to-worst range, made for projects that have not finished yet
- Cost-overflow estimate** — checked against the Ministry's own sector averages on newer projects
- Ranked watch-list** — which projects to look at this month, and why; weights set by the Ministry
- English / Hindi brief** — every sentence points to its source record; runs offline; open-source

Innovation and uniqueness of the solution

Today	With PRAKALP-DRISHTI
Promised dates	→ Calibrated window
AI assumed to help	→ AI measured (+2.6% vs OLS)
1,981-row list	→ Ranked, with reasons

made for unfinished projects · re-runnable (slide 6)



TECHNICAL APPROACH



How the numbers are made — and how we checked them

Technologies to be used (e.g. programming languages, frameworks, hardware)

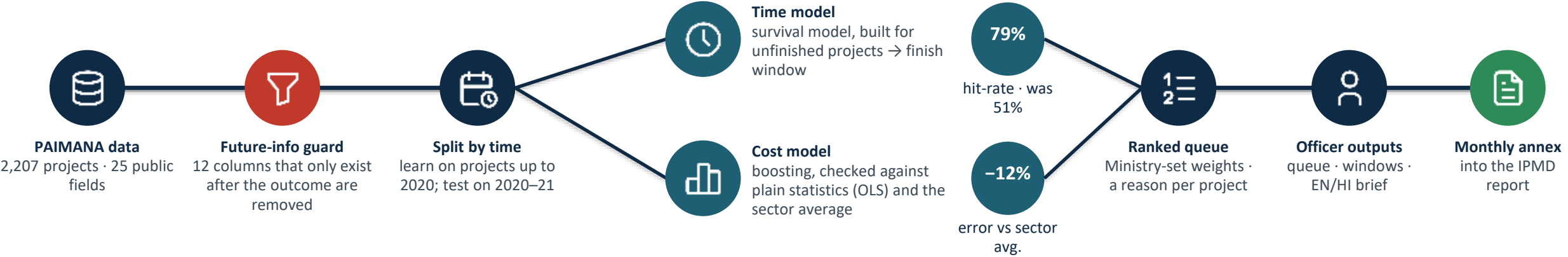
 **Python · scikit-learn · scipy**
the models and the statistics

 **FastAPI + PostgreSQL**
the service and the records

 **React**
the officer's screens

 **Docker on one NIC Cloud VM**
open-source; no licences; no external calls

Methodology and process for implementation (Flow Charts/Images/ working prototype)



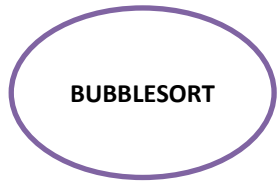
 **How we checked**
tested on projects newer than the ones we learned from · 12 future columns removed · compared with naive, sector-average and OLS forecasts · finish window checked on 596 held-out projects

 **Same result every time**
fixed seeds · identical output on any machine · 152 automated checks pass

 **What we found, honestly**
AI beats plain statistics by only 2.6% on cost · ten extra variables did not help · for time, no regression was good enough — the survival model is used


the ranked queue

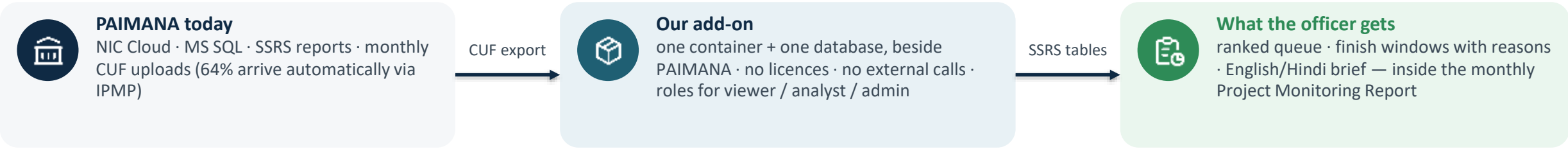

project 617185's window



FEASIBILITY AND VIABILITY



Fits beside PAIMANA as it is — and the four limits we state up front



NIC on PAIMANA's next step (Oct 2025): “AI-driven forecasting... to predict time and cost overruns, resource needs, and potential risks.”

What we need from MoSPI: actual completion dates from OCMS · the monthly archive · the fields the literature names — land status, clearance dates, milestones (untested by us)

Analysis of the feasibility of the idea

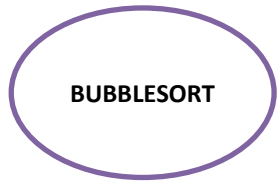
- Built** — working prototype: 20 screens · ~93 API endpoints · 152 automated checks · Docker image
- Remaining** — monthly data feed from the Flash Report archive · back-test of the queue · recalibration on real completion dates · NIC sign-on · Hindi review
- Smallest pilot** — one ministry's projects, one quarter · used by IPMD monitoring officers · owner: IPMD
- Runs on** — one NIC Cloud VM · open-source · outputs as plain tables · roles enforced server-side
- What we refuse to claim:** sub-metre satellite verification · fraud verdicts — the 20% CCEA bunching test came back null ($p = 0.68$) and we report it as null

Potential challenges and risks

- Only one snapshot of data, no month-by-month history yet — so the windows are wide (about 8 years on average)
- The queue's weights are set by policy, not yet tested against what happened next
- The cost model was tested on projects sanctioned up to 2021; 55% of the portfolio is newer
- Inputs are self-reported; some fields are missing

Strategies for overcoming these challenges

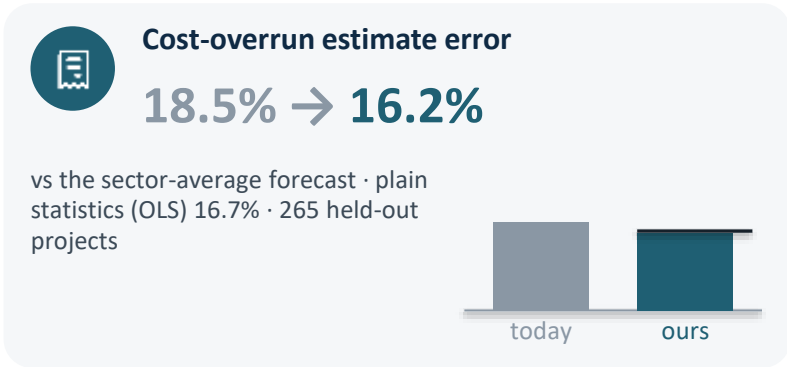
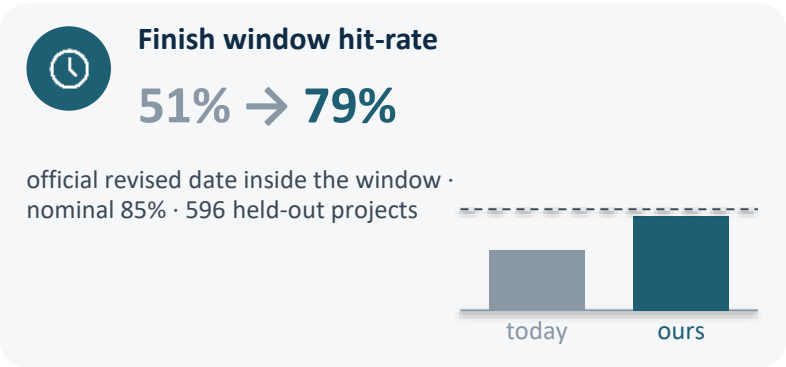
- read the Monthly Flash Report archive; add month-to-month change; same checks
- back-test now; Ministry sets the weights on its own screen
- newer projects fall back to the sector average and are labelled so — never scored as safe
- a validation check; missing signals shown as “not reported”, never as zero risk



IMPACT AND BENEFITS



From describing delays to catching them early — measured, not promised



What the officer gets — real screens from the prototype (16 Sep 2026)

Today: a 1,981-row report

A ranked list, with reasons

An escalation note, EN / HI



Potential impact on the target audience

- IPMD monitoring officers** — which projects to escalate this month, and why: computed, not compiled
- Line ministries and PMG cells** — a finish window and reasons per project, instead of a promised date
- DIID statisticians** — every number can be re-run; models, baselines and limits are written down

Benefits of the solution (social, economic, environmental, etc.)

- Governance** — ranked, reasoned, contestable; every figure has an audit trail
- Economic** — earlier intervention on a ₹42.78 lakh crore portfolio; the size of any saving is not claimed
- The pilot will measure** — officer review time per month · escalations raised before the slip · window hit-rate against real completions
- Social** — English / Hindi briefs (Hindi templates, pending official review); plain-language reasons
- Environmental** — no claim



RESEARCH AND REFERENCES



What we read — and how anyone can re-run our numbers

Details / Links of the reference and research work

on this database → the methods we used → the Ministry's own documents

Ram Singh (2010)
894 MoSPI projects, 1992–2009: delays and overruns
[LINK](#)

Morris (1990)
EPW — early evidence on public-sector overruns

Flyvbjerg (2006)
reference-class forecasting — our baseline
[LINK](#)

Wei (1992)
the survival model for projects not yet finished
[LINK](#)

Romano et al. (2019)
conformal calibration — checking the window's hit-rate
[LINK](#)

McCrary (2008)
bunching test at the 20% CCEA threshold (null result)
[LINK](#)

NIC · PIB · MoSPI
PAIMANA write-up (Oct 2025) · PIB release (Mar 2026) · April 2026 report
[LINK](#)

RERUN — three commands, every number on these slides comes back

github.com/amey-bauchkar/Prakalp-Drishti-CLAIMS.md — fitted / heuristic / refused · corrections on record (93.3% coverage and $p < 0.001$ withdrawn)

```
python analytics_engine/aft_survival.py
python analytics_engine/conformal_calibration.py
python tests/test_engine_precision.py
```

artifacts/aft_survival.json · conformal_calibration.json · overrun_models.json · seed 42 · same output on any machine

The working prototype — real screens, 16 Sep 2026

early-warning queue

project 617185

English / Hindi brief

Live ▶ prakalp-drishti.vercel.app Code ▶ github.com/amey-bauchkar/Prakalp-Drishti-CLAIMS